

A critical review of quantifying water-energy-food nexus interactions

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ABSTRACT

The water-energy-food (WEF) nexus exemplifies the holistic, integrated, and interdisciplinary system approach, emphasizing mutual interactions between water, energy, and food resources. This study presents the first systematic review that critically evaluates the past progress of WEF nexus research from the evidence-based lens of resource interactions, in terms of quantifying the types and the number of interactions studied. Using a novel classification, this review first classifies 834 WEF nexus interaction studies into four classes based on two axes: (1) The type of study (theoretical or empirical) and (2) The extent of resource interactions examined (all six interactions as a 'full' WEF nexus study or fewer interactions as a 'partial' WEF nexus study). Despite the proliferation of WEF nexus studies since 2011, no significant progress has been made towards including more resource interactions over time in either theoretical studies or empirical applications. Moreover, this review shows that: the number of resource interactions examined in empirical applications is much lower than in theoretical studies; the study of physical resource interactions remains dominant; environmental considerations are biased towards water quality and carbon emissions; and there is a misalignment between research questions and empirical methodologies. Further, this review identifies future directions and provides concrete recommendations for aligning future research to jointly achieve Sustainable Development Goals (SDGs) 2, 6, and 7.

1. Introduction

Realizing the harmonious coexistence of humans and nature through sustainable development is undoubtedly a major issue facing humankind in the 21st century. In the 2015 United Nations report "Transforming our world: the 2030 agenda for sustainable development", the seventeen Sustainable Development Goals (SDGs) were formulated to address the most pressing sustainability challenges [1]. Three of those SDGs point directly to food, water, and energy resources, i.e., SDG 2 (zero hunger), SDG 6 (clean water and sanitation), and SDG 7 (affordable and clean energy) respectively. Driven by accelerating urbanization, globalization, and climate change, the inextricable interrelationships between these three resources continue to become more pronounced [2]. Differences in the spatial and temporal distribution of resources add an additional layer of complexity to the already complex problem [3].

One of the ways in which academics and policymakers have attempted to address this complexity is by adopting a system approach [4,5]. Holistic, integrated, and interdisciplinary systems thinking is

considered a new paradigm for achieving global sustainable development [6]. As early as January 2011, the nexus among water, energy, and food was defined at the annual meeting of the World Economic Forum [7]. Later, in November of the same year, the Water-Energy-Food (WEF) nexus approach was proposed in a background work for the Bonn2011 conference. This approach was proposed as an approach that can enhance water, energy, and food security by increasing efficiency, reducing trade-offs, building synergies, and improving governance across sectors [2]. The WEF nexus proposed is an example of the system approach [5,8], which can address direct or indirect influences on the achievement of all SDGs by resource interdependencies and cascading effects beyond water (SDG 6), energy (SDG 7), and food (SDG 2) [9,10].

According to the initial definition [2], the WEF nexus emphasizes the inherent interrelations between water, energy, and food, and provides the analytical framework to reduce trade-offs, build synergies, and improve the integrated governance of resources. The consensus of WEF nexus' essence lies in two aspects: (1) A holistic perspective of the nexus [8,11,12], i.e., shifting from silo thinking to system integration, a nexus focuses on externalities across sectors, space, and time, and seeks opportunities for improving overall resource use efficiency for equal

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Abbreviations

SDGs	Sustainable Development Goals
WEF	Water-Energy-Food

benefit sharing, rather than on the productivity of isolated resources; and (2) Mutual interactions between resources [8,13–15], i.e., given the inextricable interrelations between natural resources, the nexus approach aims to capture and clarify the two-way interactions between resources. Considering the resource flows throughout supply chains, which encompass production, processing, distribution, consumption, and management activities [5,16,17], six resource interactions between three resources can be identified [13], namely.

- Energy for water: energy inputs into the water system and the water-related impacts caused by the energy system.
- Energy for food: energy inputs into the food system and the food-related impacts caused by the energy system.
- Water for food: water inputs into the food system and the food-related impacts caused by the water system.
- Water for energy: water inputs into the energy system and the energy-related impacts caused by the water system.
- Food for energy: food inputs into the energy system and the energy-related impacts caused by the food system.
- Food for water: water-related impacts caused by the food system.

Most reviews [12,18–21] on the WEF nexus provide a knowledge base of frameworks and methods, while some [22–24] provide an overview of the evolution of the nexus discipline and research. Several reviews explore the connections between WEF nexus and exogenous factors, such as pandemics [25] and climate change [26]. Some extend nexus frameworks by including more elements, such as land [16], forest [27], ecosystem service [28], biodiversity [29], and human and ecosystem health [30].

However, to date, no studies have systematically evaluated the progress of WEF nexus research from the perspective of quantifying the types and the number of resource interactions included in WEF nexus studies. This review aims to conduct a comprehensive, systematic, and extensive review of the progress of WEF nexus research from the evidence-based lens of resource interactions, in terms of quantifying which interactions are studied and how many interactions are studied. It identifies research gaps and provides perspectives on future directions. The novel contributions of this review are summarized as follows.

- The first review that systematically evaluates the progress of WEF nexus research from an evidence-based lens of resource interactions.
- A pioneering effort in synthesizing the largest database (834 studies) of WEF nexus studies, with a focus on resource interactions.
- A novel classification of WEF nexus studies, based on the type of study (theoretical or empirical) and the extent of resource interactions examined.
- Nine critical and novel findings, providing concrete recommendations on how future research could more closely align to jointly addressing SDG 2 (zero hunger), SDG 6 (clean water and sanitation), and SDG 7 (affordable and clean energy).

The subsequent sections of this review are structured as follows: Section 2.0 explains the methods used in this systematic review. Section 3.0 presents descriptive statistics, which Section 4.0 builds upon to highlight nine critical findings. Section 5.0 integrates insights from Sections 3.0 and 4.0 to outline future directions. Finally, Section 6.0 summarizes key findings and discusses limitations and potential source of errors.

2. Methods

In this study, a systematic review was conducted in accordance with the preferred reporting items for systematic reviews and meta-analyses guidelines [31]. Fig. 1 illustrates the detailed workflow of the systematic review process.

2.1. Information source

Searches on Scopus and Web of Science were performed to identify relevant scientific contributions. In addition, a backwards snowball sampling was conducted to supplement 13 missing studies neglected in the database searches. In the backwards snowballing, this study identified the candidate studies that were cited within studies included in the database search.

2.2. Search strategy

The terms “water-energy-food nexus”, “WEF nexus”, and “resource nexus” were used in the initial search in the abstract, title, and keyword. A follow-up search incorporated “food-energy-water nexus” and “FEW nexus” across the same fields. Both searches were limited to peer-reviewed studies and book chapters published in English from January 01, 2011 to August 17, 2022. The start date of 2011 was selected because it marks the year when the WEF nexus was formally defined. The end date of August 17, 2022, was chosen as it was the date of the initial search.

The limitations of the search strategy are as follows: (1) Limited search terms. The use of a limited set of keywords may restrict the comprehensiveness of the search results. Expanding search terms could yield more extensive results. (2) Language restriction. The review was limited to studies published in English. Including studies in other languages could broaden the database and mitigate language bias. (3) Exclusion of grey literature. The review focused on peer-reviewed journal papers and book chapters, excluding grey literature (e.g., conference papers, working papers, reports, and theses). Incorporating grey literature studies could strengthen the conclusions.

2.3. Eligibility criterion

The search yielded 6756 entries after deleting the duplicate items. Though all included studies explicitly use the search terms in their abstract, title or keyword, some studies do not deal with interactions between water, energy, and food resources, and some simply mention “WEF nexus” but do not delve into them. To accurately reflect the development of resource interactions in the WEF nexus research, this review only included studies with a focus on resource interactions within the nexus (referred to as ‘WEF nexus interaction studies’ in this review). Specifically, this study kept the ones that explicitly stated that the interactions between resources are part of their research aim or objective. This was implemented by screening abstracts and titles, which led to the exclusion of 5219 studies. Further, this study excluded: 22 studies published in a language other than English; 104 conference papers; and 577 studies of which full-text versions are not available online. A final dataset including 834 studies was constructed, which is the biggest database among WEF nexus reviews.

2.4. Classification

In this review, a WEF nexus interaction study that articulates the WEF nexus interactions under a holistic system thinking without giving sufficient empirical insights is classified as a theoretical study. If the WEF interactions are underpinned by sufficient empirical data and analysis, this study is classified as an empirical application.

Based on the type of study and the extent of resource interactions examined, all 834 studies were manually identified into four classes

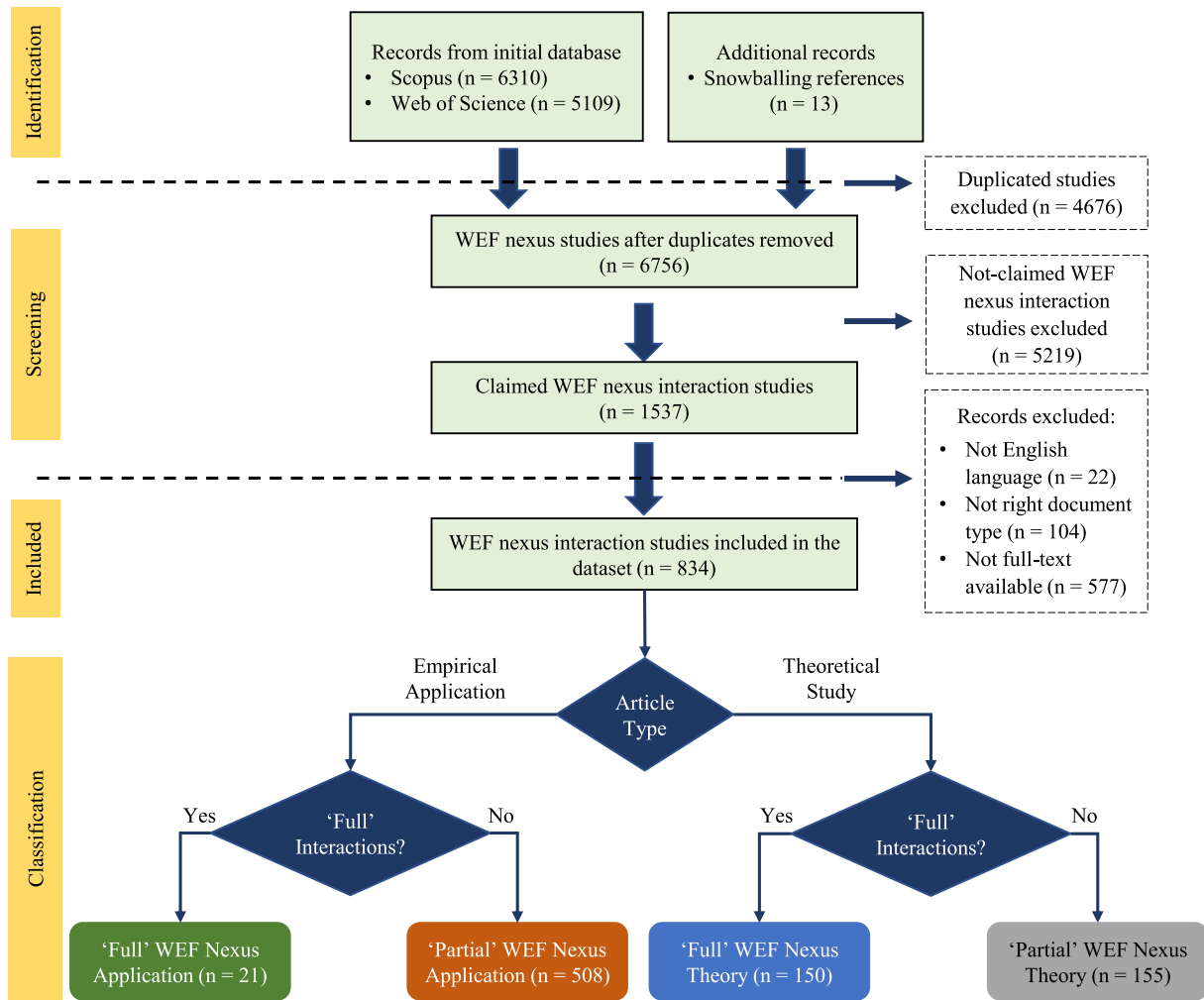


Fig. 1. Preferred reporting items for systematic reviews and meta-analyses guidelines workflow for the systematic review process. The yellow boxes in the left column show the four phases of the systematic review, the middle and right boxes respectively represent the number of studies included or excluded in each phase. The database of 834 studies can be found in: Li, W. (2024). *WEF-nexus-review-database*. GitHub repository. Retrieved July 27, 2024, from <https://github.com/RebeccaSlw/WEF-nexus-review-database>. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

based on two classifiers: (1) The type of study (theoretical or empirical); and (2) The extent of resource interactions examined (all six interactions as a 'full' WEF nexus study or fewer interactions as a 'partial' WEF nexus

study). Fig. 2 shows the classification of WEF nexus interaction studies. The distinctions between these four classes help to identify the past progress in the number of resource interactions examined in WEF nexus research.

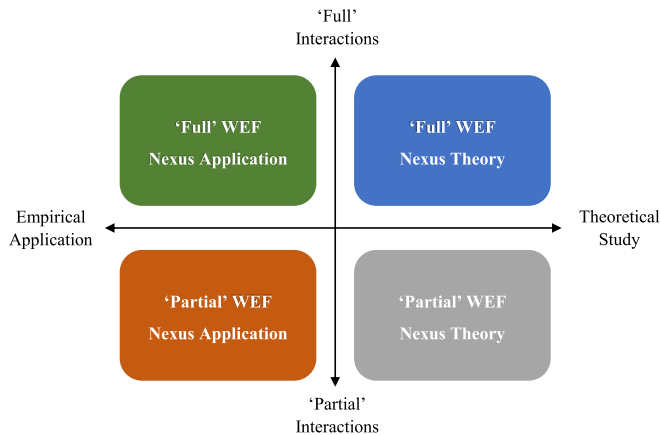


Fig. 2. Classification of WEF nexus interaction studies. The colors in this figure correspond to those in the last row of Fig. 1. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

- 'Full' WEF nexus application: an empirical application that examines all six resource interactions.
- 'Partial' WEF nexus application: an empirical application that examines several of the six resource interactions.
- 'Full' WEF nexus theory: a theoretical study that examines all six resource interactions.
- 'Partial' WEF nexus theory: a theoretical study that examines several of the six resource interactions.

3. Descriptive statistics

3.1. Shift of research focus throughout time

Fig. 3 shows the growth of WEF nexus interaction studies from 2011 to 2022. As shown in Fig. 3, since the WEF nexus concept was first put forward formally in 2011, the number of studies on this topic has grown rapidly. Until 2017, there were similar numbers of empirical applications and theoretical studies (except 2016, when theoretical studies were more than twice as prevalent as empirical applications). Since

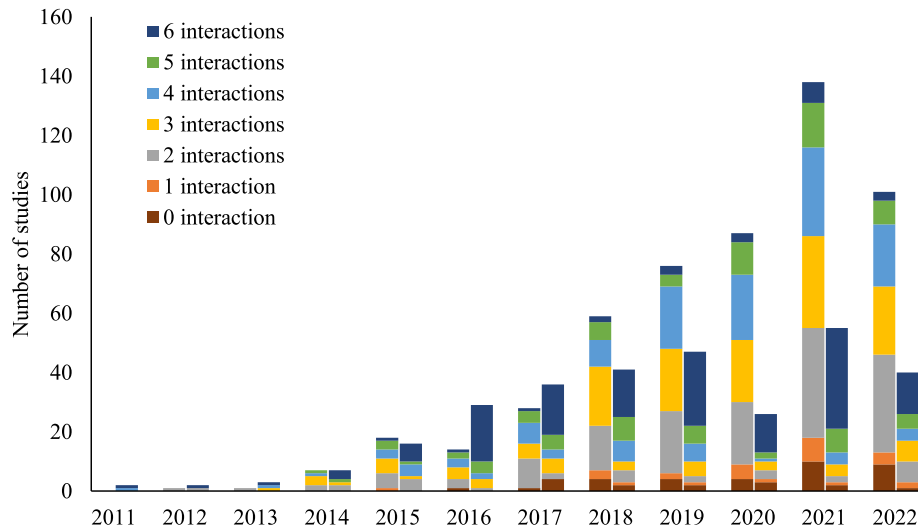


Fig. 3. WEF nexus interaction studies from 2011 to 2022. In each year, the left column denotes empirical applications, and the right column denotes theoretical studies. The data used to represent year 2022 only cover the period up to and including 17 August, and this holds for all data of 2022 in this review.

2018, the number of empirical applications has far surpassed theoretical studies, with a year-by-year increase in the size of the gap. In contrast to the proportion of ‘partial’ WEF nexus theory, the proportion of ‘full’ WEF nexus theory fluctuates distinctively over time. There are no obvious trends in the relative proportion of ‘full’ WEF nexus empirical applications to ‘partial’ WEF nexus empirical applications.

3.2. Number of resource interactions examined

The final dataset consists of 530 empirical applications (64 % of the total) and 304 theoretical studies (36 % of the total). Fig. 4 shows the number of WEF nexus interaction studies with different number of resource interactions examined. As shown in Fig. 4, only 21 % of studies ($n = 171$) examine all six resource interactions, while the remaining studies only concern partial resource interactions. There are distinct differences in the number of resource interactions examined between the two classes of studies. Almost half of the theoretical studies ($n = 150$) examine all six resource interactions, while only 4 % of the empirical applications ($n = 21$) do this, and 75 % of empirical

applications ($n = 399$) only examine between two and four interactions.

3.3. Prevalent resource interactions and interaction directions

Fig. 5 shows the six possible resource interactions in all 834 studies. In empirical applications, the differences in the attention paid to different resource interactions are more obvious than in theoretical studies. In addition, theoretical studies are more balanced in mutual resource interactions than the empirical applications. Among three sets of mutual resource interactions, the interrelation between water and food is mostly viewed from the perspective of water in both empirical applications and theoretical studies. The 84 % of empirical applications ($n = 446$) elaborate on “water for food”, while only 17 % of them ($n = 91$) consider “food for water”. Besides, empirical applications put more emphasis on “energy for food” ($n = 311$) than “food for energy” ($n = 182$).

3.4. Research scale and spatial distribution

Out of 834 studies in the dataset, 131 studies do not indicate actual research regions, including 19 studies that give the specific scale but without the actual region. Based on the remaining 703 studies that explicitly state information on both the geographical scale and the study area, the research scales of WEF nexus interaction studies are

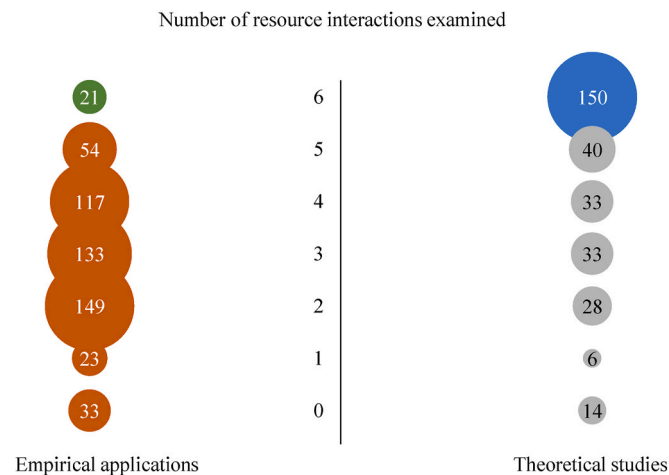


Fig. 4. WEF nexus interaction studies with different number of resource interactions examined. Numbers inside bubbles represents the number of studies. The data in this figure are all studies from 2011 to 2022. The colors in this figure correspond to the classification of WEF nexus interaction studies in Fig. 2. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

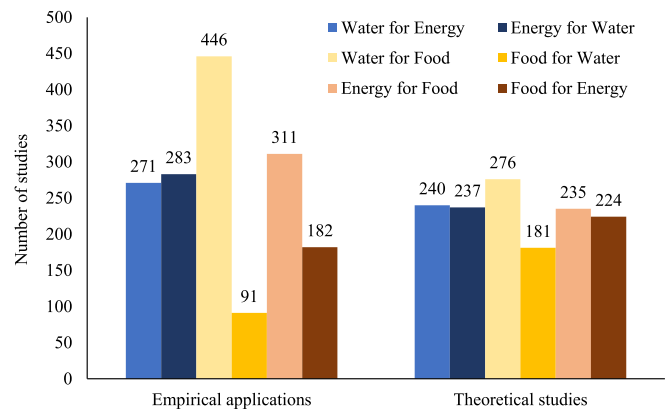


Fig. 5. WEF nexus interaction studies involving different resource interactions. The data in this figure are all selected studies from 2011 to 2022. A study can be counted multiple times if it examines several interactions.

categorized into: (1) The local scale (18 %, $n = 125$), including household [32,33], company [34], and small geographical areas, e.g., town [35], community [36], valley [37], and plain [38]; (2) The mesoscale (50 %, $n = 348$), including larger areas, e.g., city [39], district [40], state [41], province [42], region [43,44], and up to the country level [45,46]; (3) The macroscale (23 %, $n = 161$), including transboundary river basin [47] ($n = 102$) and multi-country studies [48] ($n = 59$). Without differentiating between the different parts of the same river basin, the studies on different reaches of the same river were aggregated into one river basin. Though many studies claim to be multi-country scale, in fact they only compare several countries, without analysing the resource interactions between these countries or taking all the countries as a whole. These studies are classified as country scale, which belongs to the mesoscale instead of the macroscale. The same principles are applied for determining other scales; and (4) The global scale [49] (6 %, $n = 41$). While most studies focus on one specific scale, 4 % of studies ($n = 28$) explicitly analyse multiple scales [50,51].

In addition, the spatial distribution of WEF nexus interaction studies is highly imbalanced. China receives the most attention with 130 studies having only China or regions of China as their research areas, accounting for 18 % of the 703 studies. From 2011 to 2017, the number of WEF nexus interaction studies on China was zero or one per year. However, from 2018 onwards, the number of studies on China increased rapidly, with 9 studies in 2018, 15 in 2019, 23 in 2020, 37 in 2021, and 42 until August in 2022. Of all these studies, 18 are at local scale, 91 are at the mesoscale; 18 at the river basin scale; and 3 studies consider multiple scales. Regions with the highest number of WEF nexus interaction studies at different scales are.

- City scale: Beijing (in China) ($n = 8$).
- Province scale: Jiangsu province (in China) ($n = 5$).
- Region scale: Yangtze River Economic Belt (in China) ($n = 4$).
- Country scale: China ($n = 39$), Qatar ($n = 15$), Brazil ($n = 14$), United States ($n = 13$), United Kingdom ($n = 7$), Saudi Arabia ($n = 6$), South Africa ($n = 6$), India ($n = 6$).
- River basin scale: Mekong River basin ($n = 13$), Nile River basin ($n = 9$), Indus River basin ($n = 7$), Yellow River basin ($n = 6$), Yangtze River basin ($n = 5$).
- Multi-country scale: Central Asia ($n = 8$), Mediterranean region ($n = 5$), Middle East ($n = 5$).

4. Review findings

4.1. Differences in the number and scale of WEF nexus interaction studies under specific national and regional contexts

Direct comparison of WEF nexus interaction studies across different countries can be biased due to variations in resource endowment and development levels. For example, China, Brazil, and United States, which rank in the top four in terms of the number of WEF nexus interaction studies, are among the world's ten largest countries by natural resources, area, population, and GDP [52]. One exception is Qatar. Due to water scarcity, natural gas abundance, and overdependence on food imports, there is an abundance of nexus research in Qatar [53], which ranks second in terms of number of publications at country scale in the database.

The only country with an obvious temporal trend in the number of publications is China, for which the number of WEF nexus interaction studies has increased almost five times between 2018 and 2022. This is largely driven by the central government's policy priority. In 2017, as one of fourteen key projects of China's national key research and development plan, the "Efficient development and utilization of water resources" project specified the need to develop core technologies that enable the coordinated security of water, energy, and food [54]. Given the strong dependence on natural resources, significant regional variation, fragmented institutions and governing laws, the nexus research on

Brazil is still in its infancy, with nearly half of the studies striving to explore possible resource interactions [46] and analytical frameworks at different governance scales [55]. This contrasts with other countries with large numbers of WEF nexus interaction studies, where empirical applications tend to dominate. Studies on Africa are mainly at transboundary river basin scale or multi-country scale, due to the water conflicts among countries sharing the same river basin.

4.2. Differences in the number of resource interactions examined between empirical applications and theoretical studies

From Fig. 4, the number of resource interactions examined varies between empirical applications and theoretical studies, indicating a divergence between the 'full' WEF nexus theory and 'partial' WEF nexus application. The divergence is underestimated because this review omits the conceptual frameworks constructed in the empirical applications. Theoretical studies generally come up with nexus frameworks, with almost half covering all six resource interactions [56–58]. Most empirical applications describe all six resource interactions in their theoretical frameworks. However, in the later empirical modelling, several resource interactions are directly ignored [59]. While most of them do not mention the missing resource interactions, some [60–62] explicitly acknowledge the neglected interactions, usually food for water and food for energy [50,63]. The main reasons they give is unavailable data [64], and the negligible interaction [13].

Besides, through reviewing empirical applications from diverse disciplines, it is found that hydrology and engineering related studies mainly focus on energy-water interactions [65] while economists look at overall resource allocation efficiency and welfare [62], and studies on environmental management are concerned with water footprints, energy footprints, and carbon footprints of the nexus [66]. This implies that disciplinary backgrounds of the empirical studies (e.g., social science, natural science, and engineering) may also cause cognitive bias (e.g., disparate concepts, methods, data, and highlighted resource interactions) [15,67], which leads to the 'partial' WEF nexus applications. In addition, although the number of WEF nexus studies has grown since 2011, no significant progress has been made towards including more resource interactions in both theoretical studies and empirical applications, as shown in Fig. 3.

4.3. Specific and systematic exemplification of resource interactions in application contrasts with the broad and random conceptualization in theory

Theoretical studies often unfold with the broad conceptualization of resource interactions, supplemented with concrete examples to illustrate the interactions. In empirical applications, the examination of resource interactions is usually carried out through specific examples. The analysis and examination of resource interactions in both empirical applications and theoretical studies usually narrow down to typical examples. Often-mentioned examples include agricultural irrigation [68] (water for food), virtual water transfer through food trade [26,69] (water for food), water used for fuel processing and cooling [70] (water for energy), hydropower generation [65] (water for energy), wastewater treatment [71] (energy for water), industrial waste water pollution [43] (energy for water), fertilizers usage [72] (energy for food), energy used in food transportation [36] (energy for food), conversion of food waste to energy [72] (food for energy), bioenergy production [71] (food for energy), and agricultural water pollution [68] (food for water).

In most theoretical studies, examples of resource interactions are chosen randomly due to the lack of factual context to guide the selection. However, in empirical applications, the factual contexts systematically connect resource interaction examples, which leads to several prevalent combinations of resource interaction examples. For instance, empirical applications on water allocation along a river usually involve the trade-offs between hydropower generation (water for energy) in the upstream

and agricultural irrigation (water for food) in the downstream [73,74]. Empirical applications [75,76] on the competition between energy crops and food crops integrate some resource interactions along the supply chain, including the electricity consumption for pumping underground water (energy for water), water consumption for agricultural irrigation (water for food), fertilizer and machinery application for agriculture (energy for food), energy crop for biofuel production (food for energy), and eutrophication cause by excess fertilizer (food for water).

4.4. Physical resource interactions dominate while economic resource interactions increase

Resource interactions in both empirical applications and theoretical studies are mainly derived from physical causality, embodied in the material flows from the supply to the final demand terminal [17,77]. From the factual considerations, some empirical applications [42,48,70,78–80] have gradually extended the resource interactions from the economic perspective. Examples include water demand triggered by sugar crops expansion [48] (energy for water), the effect of seasonal fallow schemes on energy consumption savings [80] (food for energy), and the impact of biofuel on food security [79] (energy for food).

The direction of economic resource interaction depends on the lens used in the specific research. Even the same physical resource interaction can be interpreted into an economic one with the opposite direction. For example, the physical water for food interaction can be interpreted as an economic food for water interaction which reflects the pressure on water caused by increased demand for agricultural products [81]. The processes within the WEF nexus are simultaneously (bio) physical, socio-economic, ecological, and political [15,19,53,82,83]. However, the interrelated interactions among these processes have not yet been properly delineated and mapped [9,23].

4.5. Biased environmental considerations towards water quality within the WEF nexus and carbon emissions outside the WEF nexus

Considering the water pollution (e.g. excessive phosphorus and nitrogen) caused by the food and energy system, the environmental impact in most theoretical studies centres on the water quality within the nexus [84]. However, interactions related to water quality are the least studied compared to other resource interactions. As shown in Fig. 5, the food for water interaction is the least examined in both theoretical studies and empirical applications. Though 53 % of empirical applications and 78 % of theoretical studies examine the energy for water interaction, most of them focus on the energy inputs into the water system rather than energy-related impacts caused by the water system.

Due to the extensive carbon emissions during energy use and agricultural production, carbon emissions are another dominant environmental consideration in most empirical applications [70,85,86]. This emphasis on carbon emissions is considered an environmental externality outside the WEF nexus.

An empirical application in Taiwan [17] stands out from most studies with a single environmental impact by adopting the ReCiPe 2008 [87] method to conduct a systematic examination of environmental impacts associated with the nexus. The ReCiPe 2008, an implementation of the life cycle assessment method, comprises eighteen impact categories at the midpoint level (including freshwater eutrophication, agricultural land occupation, fossil fuel depletion, climate change) and three aggregated endpoint categories (i.e., damage to human health, damage to ecosystem diversity, and damage to resource availability), which entails inclusive environmental impacts both within and outside the nexus.

4.6. Indirect quantification of resource interactions outweighs direct quantification, and balanced ones are scarce

The resource interactions elaborated in theoretical studies are direct

interrelations between resources. In empirical applications, some include direct indicators highlighting the interactions between the three resources, such as water for irrigation [43,88] which represents the water for food interaction, and agricultural residues used for energy purpose [72], which describes the food for energy interaction. Some empirical applications rely on indicators that are specifically related to single resources (e.g., undernourishment related to food, basic drinking water related to water, and access to electricity related to energy), and indirectly quantify the resource interactions. Some of them [89,90] employ the numerical characteristics of the indicators' scores, but they usually fail to elucidate the underlying interaction mechanism. Others use the correlation matrix to analyse trade-offs and synergies of pairwise resources [56,91,92], and the relative importance of resource interactions is usually assigned based on expert opinions. Give the symmetry properties of matrices, resource interactions examined are usually single-directional and the reverse resource interactions in the pairs are neglected [86,93,94].

Only two studies [95,96] in the dataset balance the direct and indirect resource interactions. For example, in assessing the sustainability performance of the WEF nexus in Jiangsu province, China [96], an integrated framework is constructed. This framework considers both the individual WEF resources and their interlinkages. It includes nine indirect indicators with three indicators for each resource, and six direct interlinkage indicators that evenly cover all resource interactions.

4.7. Quantification of resource interactions among regions increases while the cross-scale quantification remains deficient

To further realize the potential of nexus approaches, the consensus in theoretical studies is to expand nexus frameworks by incorporating more resource interactions across regions and scales [9,16]. Empirical applications on resource interactions among regions, which are usually interpreted to geographic externalities or spillover effects among regions, are increasing [64,82,94]. The new integrated framework of meta coupling accounts for socioeconomic and environmental interactions within and across borders, and provides grounds for characterizing resource interactions among regions and measuring spillover effects [97]. Based on the meta-coupling framework, Xu et al. conduct the first assessment about the trade-offs between regions by measuring the impacts of irrigated agriculture on the food-energy-water-CO₂ nexus across food sending systems, food receiving systems, and spillover systems.

Regarding externalities across scales, the theoretical foundation and empirical evidence are both now scarce [15]. A few preliminary qualitative explorations point out the importance of recognizing and bridging resource interactions across different scales [98], both bottom-up integration [36] and top-down integration [9]. However, most quantitative applications with multiple scales still analyse the resource interactions at different scales separately, while the cross-scale resource interactions and the mechanism behind remain unclear [99,100].

4.8. A comprehensive range of scenarios that encompass all feasible dynamics of the WEF nexus is still lacking

Resource interactions among regions and across scales are vulnerable to anthropogenic processes, such as climate change [101], urbanization [102], population growth [48], policy intervention [103], and technology advancement [104]. These factors suggests the potential variations in the demand and supply of WEF resources over time, as well as changes in resource interactions due to shifts in relative scarcity.

Several empirical applications conduct scenario simulations to illustrate the possible changes under future dynamics in resource quantity and quality (e.g., increased water demand [73], deteriorated water quality [98], changing proportion of domestically produced and imported food products [105], less meat-intensive diets, increased agriculture yields, and reduced food waste [106]), and resource interactions (e.g., biofuel expansion [48], less water-intensive crops [68],

expanded hydropower production [103]). Though several studies [73, 103] attempt to systematically construct a large ensemble of scenarios that capture the complex interactions of energy, water, and food systems, no study in the database has provided a comprehensive range of future scenarios, which accounts for all possible dynamics of resource interactions among regions and across scales (e.g., potential shocks and risks).

4.9. Misalignment of research questions and empirical methodologies

Despite the proliferated WEF nexus studies, the development of methodologies for identifying and quantifying resource interactions has not kept pace [19]. Current nexus methods predominantly adapt traditional disciplinary approaches, such as life cycle assessment and ecological footprint from environmental management [107], input-output analysis from economics [108], and game theory from mathematics [109]. While these borrowed approaches hold potential to advance WEF nexus research, they often inadequately capture mutual resource interactions and elucidate the mechanisms behind them [9,43, 110], resulting in a misalignment between research questions and empirical methodologies. This misalignment arises because these approaches often fail to adequately adapt to the holistic and interactive nature of the WEF nexus.

This misalignment can be observed in various empirical applications. For example, many empirical applications employ the ecological footprint method in combination with top-down input-output analysis [108, 111,112], bottom-up life cycle assessment [69,85,97] or data envelopment analysis [90,113], to calculate water footprints, energy footprints or carbon footprints based on the resource flows along the supply chain. In the input-output footprint analysis, the identification and quantification of resource interactions largely depends on the precision and sophistication of the metrics, and no study in the dataset examines all six resource interactions using this method. Empirical applications using the life cycle footprint method usually merely consider the single-direction interactions, determined by the consumption-based accounts of the calculation principles [114]. The data envelopment analysis still treats the resource interactions within the nexus as a “black box” and fails to identify them. Optimization models are promising methods; however, they sometimes fail to identify and examine all resource interactions within the WEF nexus [33,115]. This limitation is not inherent to the method itself but rather stems from the lack of ‘full’ consideration of the WEF nexus in its conceptualization and application.

System modelling and integrated modelling align well with the holistic and interactive nature of the WEF nexus, making them well-suited for tackling complex, interdisciplinary challenges within the WEF nexus. System modelling emphasizes interactions among WEF resources, while integrated modelling combines knowledge from multiple disciplines. These approaches, rather than specific models, provide frameworks for constructing WEF nexus models. Some studies [33,116] adopt a system approach to highlight the resource interactions within the WEF nexus, while others [63,115] employ an integrated approach to integrate models from multiple disciplines. However, integrated models often face challenges in linking various sub-models, and both integrated and system models often fail to fully identifying all interactions within the nexus due to the lack of ‘full’ consideration of the WEF nexus.

5. Future directions

5.1. Toward a ‘full’ consideration in the conceptualization and application of the WEF nexus

It is not the purpose of this review to provide a rigorous paradigm for a ‘full’ WEF nexus theory or application that constrains all nexus-related research. Instead, it emphasizes the importance of taking a ‘full’ consideration in the initial conceptualization and empirical application of WEF nexus studies. The ‘full’ consideration encompasses: (1) All

possible interactions between resources, among regions, and across scales; (2) Bridging the discrepancies of resource interactions between theoretical studies and empirical applications; and (3) A comprehensive range of feasible future dynamics of the WEF nexus.

5.2. Transformation to in-depth transdisciplinary integration in the WEF nexus

To fully realize the great potential of the WEF nexus in jointly achieving multiple SDGs, notably SDGs 2, 6, and 7, it is essential for stakeholders, policy makers, and academics across diverse disciplines (e.g., social science, natural science, and engineering) to actively engage in and substantially enhance transdisciplinary dialogues. These dialogues are crucial for facilitating the interaction and co-creation of new WEF nexus knowledge by integrating theories, methods, practices, and implementations [117]. Such integration advances theoretical understanding and support policymakers in the design and regulation of water, energy, and food systems.

Specifically, by providing frameworks that allow contributions from various disciplines and the involvement of non-academic participants, transdisciplinary integration enables a more comprehensive examination of all possible resource interactions from a wide range of causal perspectives. For policymakers, a transdisciplinary perspective allows for a thorough assessment of both direct and indirect resource interactions. Subsequently, cost-benefit analysis can be employed to secure common support and funding for interventions that benefit multiple stakeholders.

5.3. Mutual interactions between environment and the WEF nexus

Many studies pay attention to the one-way effect of the WEF nexus on the environment (e.g., CO₂, NO_x and SO₂ in waste gas from energy consumption, COD and NH₄-N in wastewater from food production). However, they often ignore the reverse effect of the discharged pollutants on the WEF nexus. An advance [83] illuminates that the reaction of environmental pollutants (i.e., waste gas used for waste heat recovery and recycled wastewater) has a significant positive effect on water saving, energy saving, and pollutant reduction. It is well established in biological sciences that the elevated CO₂ level can increase plant growth and stimulate crop yields [118], while simultaneously reducing the nutritional value of some food crops (e.g., protein, micronutrients, and vitamin content) [119]. Despite this, the carbon emissions of the WEF nexus are still viewed as environmental externalities in research. Furthermore, there is growing recognition that WEF interventions must align with and, where feasible, integrate with the ecosystem [27,120]. For such interventions and policies to be effective, emissions cannot be disregarded as external to the WEF nexus.

Additionally, empirical studies primarily examine the effects of external shocks on the WEF nexus through scenario analysis. Given the potential for vicious cycles involving natural disasters and conflicts, it is both crucial and urgent for future research to not only assess the impacts of extreme weather events on the WEF nexus but also to explore whether the WEF nexus can offer novel opportunities to mitigate and address these emerging challenges [5,18].

5.4. Realize integrated, inclusive, and equitable resource governance through the WEF nexus

A crucial transformation needed is moving from nexus thinking to nexus doing, and gradually to nexus governance [121,122]. The nexus approach seeks to achieve holistic optimization with equal benefit sharing through integrated planning, management and governance across sectors and regions, rather than transferring the problem to other resources, regions, or groups. By leveraging the integrated and inclusive nature of the WEF nexus approach, future research can be guided to jointly address SDGs 2, 6, and 7, and further contributing to achieving

all SDGs by resource interdependencies and cascading effects beyond water, energy, and food [9]. Promising examples include the focus on the WEF nexus in small island developing states, which encounter intensified sustainability challenges due to their small size, remoteness, and limited resources [123].

Additionally, future WEF nexus research should pay greater attention to equity. As urbanization, globalization, and climate change accelerate, power imbalances among sectors, regions, and scales are likely to exacerbate inequalities among various groups (e.g., by gender [124] or income [125]). Resource governance consideration of equity often benefits certain groups at the expense of others. For instance, a study [73] underscores that well-intended climate change mitigation policies introduced in wealthier countries exacerbate the vulnerabilities of the WEF nexus throughout the developing world.

6. Conclusions

This study provides a comprehensive, systematic, and extensive review of the progress of WEF nexus research through an evidence-based lens of resource interactions. It makes four novel contributions: First, it is the first systematic review to evaluate WEF nexus research by quantifying the types and the number of resource interactions included in these studies. Second, it synthesizes the largest database of WEF nexus studies, comprising 834 studies, with an emphasis on resource interactions. Third, it introduces a novel classification of WEF nexus studies by distinguishing between theoretical studies and empirical applications, and by assessing the extent of resource interactions examined. Fourth, it identifies nine critical findings, offering concrete recommendations for how future research to better jointly address SDG 2 (zero hunger), SDG 6 (clean water and sanitation), and SDG 7 (affordable and clean energy). This review is highly relevant for policymakers in shaping research and innovation agendas, and for research by offering guidelines for new investigations aimed at effectively contributing to the joint achievement of SDGs 2, 6, and 7.

Building upon the extensive database and rigorous analysis, this review reveals nine key insights: (1) Despite the proliferation of WEF nexus studies since 2011, no significant progress has been made towards including more resource interactions over time in either theoretical studies or empirical applications; (2) The number of resource interactions examined in empirical applications is much lower than in theoretical studies; (3) In empirical applications, interactions are examined through specific and systematic examples, whereas in theoretical studies, resource interactions are conceptualized using broad and random examples; (4) The study of physical resource interactions remains dominant; (5) Environmental considerations are biased towards water quality and carbon emissions; (6) Indirect quantification of resource interactions outweighs the direct quantification, and balanced ones are scarce; (7) Cross-region quantification of resource interactions has increased while cross-scale quantification remains deficient; (8) A comprehensive range of scenarios that encompass all feasible dynamics of the WEF nexus is still lacking; and (9) There is a misalignment between research questions and empirical methodologies.

Based on these critical findings, this review identifies future research directions and provides concrete recommendations on how future research could more closely align to jointly addressing SDGs 2, 6, and 7: (1) Move towards a 'full' consideration of both conceptualization and application of the WEF nexus; (2) Transform to in-depth trans-disciplinary integration in the WEF nexus; (3) Consider mutual interactions between the environment and the WEF nexus; and (4) Realize integrated, inclusive, and equitable resource governance through the WEF nexus.

While acknowledging its contributions, this review possesses certain limitations in its searching strategy that warrant further investigation: (1) Restricted search terms, which may limit the comprehensiveness of the results. Expanding the keyword list to include additional terms could improve the breadth of the search and yield more comprehensive

results. (2) English language restriction. Including studies in other languages could broaden the database and mitigate language bias. (3) Exclusion of grey literature. The review focus on peer-reviewed journal papers and book chapters. Incorporating grey literature could strengthen the conclusions.

Potential errors in the review results may arise from manual classification and synthesization of studies, as WEF nexus research employs diverse frameworks and methodologies, complicating accurate data capture. These errors may affect result accuracy and lead to biased conclusions. To improve accuracy, future reviews could incorporate generative pre-trained transformer algorithms to assist with data classification and validation checks, thereby enhancing the precision of identification and synthesis.

CRedit authorship contribution statement

Wei Li: Conceptualization, Methodology, Formal analysis, Visualization, Writing – original draft. **Philip J. Ward:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Lia van Wesenbeeck:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors confirm that the data supporting this study are available within the article.

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